

Abstract

People often pass one another answers that took much longer to learn than they take to say. A warning can fit in a sentence; the observations, comparisons, and conditions that made the warning useful may not. Someone can hear the advice, believe it, and follow it correctly while the situation stays familiar, yet still be poorly prepared to recognize when the advice no longer applies or when the reason behind it has changed. This paper asks what has to survive when knowledge is passed from one person or system to another so that the recipient can benefit from what was learned without becoming unable to judge when the inherited answer no longer fits. Correct routine performance cannot answer that question by itself. The proposed experiment first establishes common competence at a familiar task, then varies what a fixed-dose handoff contains and tests held-out cases that require keeping a rule under irrelevant change, withholding it when the case leaves its declared domain, or re-deriving an action when a parameter in the functional relation supporting the rule changes. A separate evidence-strength control checks the simplest always-preserve or always-revise strategies. The design is explicitly bounded by established transfer, preparation-for-future-learning, Cognitive Flexibility Theory, and Double Treatment assessment logic; it does not claim a new learning mechanism, acquisition-by-assessment principle, or proprietary profile model. The remaining handoff-specific composition is OPEN and should contract if simpler established designs, scoring schemes, or models are equally informative.

Keywords: knowledge transfer; adaptive expertise; conditional knowledge; conceptual change; transfer-appropriate processing; diagnostic classification models; knowledge handoff

1. Introduction

Much of what people know arrives second-hand. A parent can warn a child in a few words about something that took years to learn. A senior engineer can say, "Do not configure it that way," without replaying every failure that made the rule obvious. A friend can offer a conclusion reached through months of observation. This compression is one of the reasons human learning can accumulate at all: we do not each have to rediscover every useful lesson from scratch.

But a short answer can travel more easily than the structure that made the answer usable. A person may remember the warning, trust the person who gave it, and behave correctly while nothing important changes. What may be missing only becomes visible later: why the answer applied, where it stopped applying, which differences were harmless, and what kind of change should make the person reconsider it. That is why "I told you" can be both true and incomplete.

The other person may really have heard the words. The question is whether the words carried enough of what made the conclusion appropriately usable.

A familiar way this discrepancy appears in ordinary life is some version of: "I know you told me. I believed you. I still did not understand what you meant until now." That sentence is not evidence that direct experience is required, nor does it prove that the person giving the advice was right. It is an intuitive doorway into the scientific problem. What has to survive when knowledge is handed from one person or system to another so that the recipient gets the benefit of what was learned without losing the ability to recognize when the inherited answer no longer fits?

Psychology already owns most of the pieces needed to ask that question carefully. Work on procedural, conceptual, and conditional knowledge separates knowing what to do from knowing when and why; adaptive-expertise and preparation-for-future-learning research show that routine success can conceal differences that appear under novelty; Cognitive Flexibility Theory warns that oversimplified knowledge can become rigid when situations are complex or change; conceptual-change research studies when people preserve or revise beliefs; analogical-transfer and contrasting-case work separate superficial variation from deeper structure; assistance research asks what support changes about later independence; organizational and design-rationale research show that rules and artifacts can survive while context is lost; and social epistemology explains how knowledge can be acquired from other people (Audi, 2013; Malfatti, 2020; Marchal-Crespo & Reinkensmeyer, 2009; Paris et al., 1983; Rittle-Johnson & Alibali, 1999; Schwartz & Bransford, 1998; Schwartz et al., 2005, 2011; Schwartz & Martin, 2004; Spiro et al., 1988, 1992). Healthcare handoff research likewise treats transfer between people as a

continuity problem involving information before, during, and after the handoff (Abraham et al., 2012).

This paper does not rename those established ideas. It puts them around one practical question: when an answer is compressed for someone else, what has to remain available so that the person can keep the answer when it is still right, limit it when the situation falls outside its intended range, and change it when the reason it worked has changed? The controlled experiment proposed later is a way to make that hidden difference observable.

The proposal is therefore methodological, not a claim about a new mental system. Two people can look equally competent while everything stays familiar even though only one of them has enough usable structure for a changed situation. Later sections call this an observational-equivalence problem, but the ordinary point comes first: present success can hide a difference that matters only later. The experiment is designed to expose that difference without pretending that behavior alone tells us the unique internal route that produced it.

2. What Correct Performance Tells Us - and What It Does Not

2.1 Help can produce success without revealing what the person can do alone

Someone can look successful while help is doing more of the work than that success reveals. Koedinger and Aleven (2007) framed the assistance dilemma as the problem of how much information or help to give versus withhold in order to support learning. In a motor-learning task, Tullis et al. (2015) found that constant assistance improved training performance but produced the worst later unassisted test performance, whereas faded assistance produced the best unassisted performance. The lesson is not that help is harmful. It is that doing well with help and being able to act independently later are different measurements.

Rehabilitation robotics reaches a parallel conclusion from a control perspective. Marchal-Crespo and Reinkensmeyer (2009) reviewed assistive, challenge-based, haptic-simulation, and coaching strategies and described the motivation for assistance-as-needed: guidance can alter the task experienced by the learner and can reduce the learner's own contribution. Virtual-fixture research likewise distinguishes systems that guide a user along a desired path from systems that mainly constrain the user to a safe region (Bowyer et al., 2014). Modern safety filters provide a related engineering architecture by modifying a proposed control input when needed to satisfy safety constraints rather than replacing the nominal controller wholesale (Cohen & Belta, 2023; Wabersich & Zeilinger, 2021).

For the present question, this means that "Was the person helped?" is too crude. Support can protect, teach, and still leave room for learning. What matters is what the support changes about what the person has to generate, notice, and use. The engineering examples above are architectural analogies only: they show that constraining unsafe actions can differ from replacing a person's policy, not that human learning runs on the same mechanism.

2.2 Knowing what to do is not the same as knowing when it applies

Remembering what to do is not the same as knowing when that answer belongs. Paris et al. (1983) emphasized strategic knowledge in context: a strategy includes knowledge of when, where, and why it should be used, not only the action itself. Rittle-Johnson and Alibali (1999) experimentally separated conceptual from procedural instruction in mathematical equivalence. Procedural instruction promoted use of the taught procedure but showed limited transfer, whereas conceptual instruction supported understanding and the generation or transfer of correct procedures.

The same human distinction appears in adaptive expertise. A person may be very efficient when the familiar procedure still works and much less prepared when the familiar procedure stops being enough. A scoping review found substantial convergence around the importance of both efficiency and innovation, while also identifying continuing measurement challenges (Kua et al., 2021). Aliaga et al. (2024) likewise found no difference among training groups on familiar cases assessing routine expertise, while performance diverged on novel cases designed to assess adaptive expertise. In ordinary language: equal routine performance does not guarantee equal readiness for change.

2.3 Some differences appear only when the next situation changes

Sometimes two people look equally prepared until the next situation gives them something new to learn from. Preparation-for-future-learning research makes that possibility explicit. Schwartz and Bransford (1998) showed that working through contrasting cases can create differentiated prior knowledge that makes later telling more usable. Schwartz and Martin (2004) went further: their invention studies treated later access to learning resources as part of the assessment and showed why a standard endpoint can miss differences that become visible when learners have to learn again. Schwartz et al. (2011) likewise showed that when telling occurs can affect learning of deeper structure and later transfer. The general lesson is already established: present proficiency need not reveal how well what was learned will support later learning or transfer.

Dynamic assessment asks a similar question from another direction: instead of treating today's score as the whole story, what happens when the person is given feedback, teaching, or scaffolding? A systematic review of reading research found that dynamic measures explained variance in later growth beyond static measures in multiple domains, although effects varied

across tasks and studies (Dixon et al., 2023). For the present paper, these literatures remove novelty from the idea that routine success can hide later differences. The remaining question is specifically about what a compressed handoff supports when changed conditions demand keeping, limiting, or changing an inherited answer.

2.4 Rules can outlive the reasons that made them useful

A rule can keep being repeated after the context that once made it sensible has thinned or changed. Organizational research gives a group-level version of this problem. Feldman and Rafaeli (2002) argued that routines create understandings not only about what is done but also about the goals those routines serve, and that these understandings affect adaptation. Feldman and Pentland (2003) likewise distinguished the abstract or ostensive aspect of a routine from its concrete performances.

This matters because success yesterday can become inertia tomorrow. Engel et al. (2026) found that learning environment affected the speed with which participants adapted after a rule change, illustrating the tension between stability and flexibility. Routines are not therefore defective; they reduce uncertainty and coordination costs. The point is simply that past compliance does not show that a person or group will know when the governing conditions have changed.

Engineering makes the loss easy to picture because a finished design can outlive the reasoning that produced it. Design-rationale research distinguishes an architecture from the reasons, alternatives, assumptions, and constraints that shaped it. Bratthall et al. (2000) tested whether retrospective design rationale improved change-impact analysis; benefits in correctness and speed appeared for one of two systems, with the second yielding inconclusive results. Türic (2026) described transition-boundary failures in terms including compressed handoff and

orphaned rationale, and Zhang et al. (2026) likewise treated reusable experiential knowledge as context-qualified and subject to later updating. So preserving rationale, context, applicability boundaries, or updating pathways is not a new idea owned by this paper.

2.5 Good revision also means knowing when not to revise

Good judgment is not simply a willingness to change one's mind. A person also has to know when a surprising observation is too weak, irrelevant, or poorly grounded to justify abandoning a good rule. Conceptual-change research is close prior art here. Chinn and Brewer (1998) tested a taxonomy of responses to anomalous data that included ignoring or rejecting data, questioning their validity, excluding them from a theory's domain, holding them in abeyance, reinterpreting them, making peripheral changes, and changing theories. Hemmerich et al. (2016) later showed that confidence change and explicit theory change depend on the structure of the evidence. None of that preserve-versus-revise behavior is new here.

Likewise, psychology already knows that a change in appearance is not the same as a change in structure. Holyoak and Koh (1987) showed that both surface and structural similarity affect analogical transfer, with structural similarity becoming especially important for successful use once relevance is recognized. The present paper therefore does not claim the surface-versus-structure distinction. Its narrower question is whether differences in what survived a handoff can be made visible by asking people to respond to several kinds of change after everyone first looks competent in the familiar case.

2.6 Established transfer and assessment research already explains much of this

The strongest conventional explanation is transfer-appropriate processing: later performance improves when what was learned matches what the later task requires. Morris et al.

(1977) showed that the benefit of an encoding operation depends on the later test demands, so selective handoff-content by later-demand interactions need no new handoff construct.

Instructional-transfer studies are even closer. Perry (1991) found comparable learning but different transfer after principle- versus procedure-based instruction. VanderStoep and Seifert (1994) found that instruction about when principles apply improved principle selection, and Eiriksdottir and Catrambone (2011) review how procedures, principles, examples, and transfer goals interact. Applicability instruction is therefore established science.

Explanation is also established. Barnett and Ceci (2002) reject transfer as a single undifferentiated quantity. In worked-example research, adding process information did not always improve transfer and could become redundant (van Gog et al., 2006, 2008). Rationale is therefore something to test, not assume.

Other explanation studies show why the test remains worthwhile. Explanation can promote broader generalization or attention to causally informative properties (Walker et al., 2014; Williams & Lombrozo, 2010), integrated conceptual "why" information can support novel-task transfer (Cheung et al., 2019), and explanation can affect hypothesis generation without improving evaluation (Brockbank & Walker, 2022). These selective effects areprecedented.

Cognitive Flexibility Theory already addresses how oversimplified representations can become rigid in complex or changing domains and why adaptive performance may require reorganizing knowledge across cases and contexts (Spiro et al., 1988, 1992, 2019). This paper therefore does not claim that compressed knowledge becoming brittle under novelty is new.

Preparation-for-future-learning work similarly shows that ordinary endpoint performance can miss differences that appear under later learning or assessment (Schwartz & Bransford,

1998; Schwartz et al., 2011; Schwartz & Martin, 2004). Halbherr and Kapur's (2020) Double Treatment design explicitly varies both learning and assessment treatments to expose acquisition-by-assessment interactions. That higher-level crossing is established.

What remains OPEN is narrower: routine competence is established before treatment, the content of a compressed post-competence handoff is varied, and independently generated held-out cases require preservation, boundary restriction, or rationale-sensitive re-derivation. Evidence strength remains a separate control, and the design must contract if simpler scoring, factorization, or established analyses are equally informative. A bounded hostile search found no single prior framework clearly containing that full combination; that does not establish novelty.

Contrasting-cases research also owns the logic of changing key features while holding irrelevant ones fixed. Schwartz and Bransford (1998), Chase et al. (2019), and Loibl et al. (2020) provide relevant precedents, although Loibl et al.'s contrasting-case manipulation did not improve posttest outcomes. The present proposal uses such contrasts after a compressed handoff; it does not claim structural contrast as new.

2.7 Established diagnostic models already know how to score response patterns

Psychometrics already has tools for scoring multivariate response patterns. Diagnostic classification models relate item responses to multiple discrete attributes through preregistered item-to-attribute mappings, with mature approaches to model fit, Q-matrix specification, learning trajectories, and formative use (Henson et al., 2009; Rupp et al., 2010; Tatsuoka, 1983; Wang et al., 2026). That machinery is not a contribution of this paper.

Diagnostic models can also be used around interventions: Madison and Bradshaw (2018) extended a transition diagnostic classification model to multiple groups for differential attribute growth. The present paper therefore needs no new profile model. DCMs or other conventional

multivariate models may be used when appropriate; what remains OPEN is the upstream handoff-specific experimental composition, not a new scoring architecture.

3. What This Paper Is Not Saying

3.1 People do not have to live through everything themselves

This problem can tempt us into saying, "They had to live it themselves." The evidence does not support that general rule. Testimony is a central source of human knowledge (Audi, 2013), and social epistemology does not support a simple claim that understanding cannot be transmitted through testimony (Malfatti, 2020). Cumulative culture is an even clearer counterexample: useful practices can improve across people even when each person does not reconstruct the entire causal history. Zwirner and Thornton (2015), for example, found cumulative improvement across transmission chains even when teaching was unavailable and participants learned by inspecting prior products.

The aim is therefore not to force people to learn the hard way. A person's lack of first-person experience is not, by itself, an epistemic failure. If the relevant structure can be passed on well enough, that is exactly the point of social learning. The research target is smaller and more useful: what has to be transmitted so that the person can handle the future decisions they are actually likely to face?

3.2 Protection and guidance are not enemies of learning

Nor does helping or protecting someone necessarily make them less capable. Worked examples, direct instruction, coaching, scaffolding, safety filters, and assistance-as-needed can all improve learning under appropriate conditions. The useful comparison is not help versus no help. It is what different forms of help leave the person able to notice, contribute, and update.

A safe environment can still preserve opportunities to learn. Freedom can also fail to teach if the evidence needed for updating never becomes available or never gets used. So neither protection nor independence should be treated as a mechanism by itself.

3.3 The source's full history does not have to be copied

The goal is not to copy the source's life into the recipient. The source's history may include years of failed attempts, tacit comparisons, emotional salience, and local details that the recipient will never need. If a compressed explanation preserves everything required for the later decisions the recipient faces, the missing history is scientifically unnecessary for that purpose.

This is also why the intuitive phrase "feeling the weight" should not be mistaken for a mechanism. What matters may be much more ordinary and transmissible: knowing which conditions matter, which do not, what would count against the inherited answer, and what can change without invalidating it. Emotion may accompany that understanding, but the experiment below tests informational structure, not emotional replication.

3.4 There is no universal packet of information to hand over

There is no universal answer to "What should I tell them?" A stable procedure may require little more than accurate execution. A safety-critical rule may appropriately favor conservative preservation. A model expected to travel across settings may need explicit assumptions and falsifiers. A social judgment may involve different evidence and uncertainty. The content that needs to survive depends on what the person will later have to do.

That is the central boundary: enough for what? A handoff can be perfectly adequate for today's routine and inadequate for tomorrow's exception.

4. When Has Enough Survived the Handoff?

Imagine two people who were both told the same answer and both use it correctly today. Tomorrow the situation changes. One recognizes that the change is irrelevant and keeps the rule. Another recognizes that the case has crossed the rule's boundary and stops using it. A third kind of change leaves the case inside the boundary but changes a parameter in the functional relation that made the old action adequate, so the appropriate action must be re-derived. The practical question is whether the handoff supports those later decisions, not simply whether it produced today's correct behavior.

This paper uses the term handoff sufficiency only after that ordinary-language problem is clear. A knowledge handoff is sufficient for a particular future demand when enough relevant structure survived for the recipient to use the inherited knowledge appropriately and to revise it when the relevant conditions change. The term is functional and demand-relative, not a new psychological mechanism. It does not require the source's complete history, the source's subjective confidence, or an identical internal representation. It also includes restraint: someone who abandons a valid rule whenever anything surprising happens is not more adaptive than someone who never revises it.

**Source history → compressed handoff → recipient competence → familiar use →
changed conditions → preservation, restriction, modification, or abandonment**

The familiar-use stage in the sequence above matters because it is where the hidden difference can disappear from view. Two people may have the same accuracy, speed, confidence, or compliance while the inherited rule still works. Only when the situation changes do we discover whether both people had what they needed to tell an irrelevant change from a boundary failure or a change in the relation that supported the rule. In formal terms, familiar performance creates an observational-equivalence problem.

5. Changing the Situation on Purpose

To investigate the question, the experiment has to change the world on purpose. The first change should look different while leaving everything that makes the rule valid intact. The second should move the case outside the rule's intended domain. The third should keep the case in-domain while changing a parameter in the functional relation that made the baseline action adequate. Those contrasts themselves are not new: surface-versus-structural transfer, contrasting cases, conceptual change, and multidimensional transfer already establish much of this logic (Barnett & Ceci, 2002; Chase et al., 2019; Chinn & Brewer, 1998; Hemmerich et al., 2016; Holyoak & Koh, 1987; Loibl et al., 2020). The point of the battery is narrower. The test demands are defined from manipulated task relations before anyone responds, and the held-out cases must not simply repeat triggers, exceptions, parameter-action pairs, or answers stated in the handoff. The experiment is a structured assessment of how different handoffs support later performance, not a causal decoder and not a new form of diagnostic profiling.

Table 1
Predeclared structural probes for evaluating a knowledge handoff

What kind of change?	How the experiment defines it	What the person should do	What this asks
Surface / irrelevant change	Surface or contextual features change, but the rule's applicability boundary and supporting relation stay the same.	Keep the rule.	Can the person avoid overreacting to a change that does not matter to the rule?
Boundary change	The case moves outside a predeclared applicability boundary, while the rule's supporting relation inside its valid domain is unchanged.	Classify the case as outside the declared domain and withhold the rule.	Does the handoff support recognition that the rule no longer applies to this case?

What kind of change?	How the experiment defines it	What the person should do	What this asks
Supporting- relation parameter change	The case remains inside the applicability boundary, but a held- out parameter in a task-generating functional relation changes. The rationale states the general relation, not the critical parameter- action pair; the prompt does not state the replacement action.	Use the retained functional relation to derive and apply the modified action.	Does the rationale support a new derivation when the relation between relevant quantities changes, rather than retrieval of a taught lookup?

The three rows in Table 1 are not proposed as three natural kinds of mind. They are three ways the experiment can change the task while holding different relations fixed. In the boundary case, the old rule still has the same within-domain logic, but the new case is no longer in that domain. In the third case, the case remains in-domain, but a parameter in the functional relation that supported the baseline action changes; the correct action must be re-derived from the general relation rather than retrieved from a taught condition-action pair. Preserve, restrict, and modify are therefore task-defined responses, not a new psychological taxonomy. Evidence strength stays separate as a control: it asks whether a person can resist a weak reason to abandon a good rule

while remaining able to revise under strong evidence. Because every group receives the same evidence-state mapping, no handoff-specific advantage is predicted for that control.

A pattern across several probes can tell us more than a single total score, but it still cannot tell us uniquely why a person succeeded or failed. The same outward pattern might come from memory, comprehension, motivation, authority sensitivity, task strategy, or other causes. Stronger causal claims would require independent measures or interventions. The modest claim is functional: the battery tests whether different handoffs support appropriate responses across declared later demands and whether those response patterns survive comparison with simpler scoring and conventional models.

5.1 Established analysis models are enough

The analysis should not be allowed to manufacture a new theory out of a familiar pattern. The simplest possibility, Model A, is just "more information helps more." If richer handoffs improve everything broadly, there is no reason to invoke a selective architecture.

A stronger conventional account, Model B, says that later performance depends on whether what the person received matches what the later case requires. Handoff content can be coded in advance as rule, applicability information, and rationale; held-out items can be coded independently by the relation they change and the information or processing they demand. A scientifically adequate implementation may include preregistered handoff-by-probe interactions, participant and item heterogeneity, random slopes or analogous multivariate terms, plus justified nuisance predictors such as difficulty and surface distance. Model B is not artificially weakened to make room for a proprietary alternative.

No separate profile model survives that comparison. Established transfer models can represent selective content-demand matches, and diagnostic classification models can represent

item-to-attribute mappings and multivariate response patterns. The battery may therefore be analyzed with those or other conventional multivariate tools when their assumptions fit. Better fit by a richer model would justify that implementation, not a new psychological construct. The scientific object under test is upstream of the statistics: whether this controlled handoff-and-change design exposes a useful difference in what remained usable after the handoff.

6. Advice as a Motivating Case, Not Proof

Advice is where the problem often feels most recognizable. One person can compress months or years of observation into a sentence: "Do this," "watch out for that," or "this is the part that matters." The other person may listen carefully, trust the source, and remember the conclusion. Later they may still find themselves saying, in effect, "I know you told me. I believed you. I just did not understand what you meant until now." That experience motivates the paper because it separates hearing a conclusion from having enough of its usable structure. It does not show that personal experience is necessary, that the advisor had privileged access, or that the recipient should have obeyed.

Real advice cannot validate the experiment because ordinary relationships and life decisions rarely give us independently known ground truth or cleanly assigned changes. The framework therefore does not tell a person when to defer to advice, ignore it, leave a relationship, remain in a project, or make any other life decision. Advice is a motivating case and an application domain. The empirical burden remains with controlled tasks in which the relevant relations and appropriate responses can be specified before the participant acts.

7. A Controlled Experiment for the Question

The experiment is built to answer the human question without pretending that ordinary life supplies clean measurements. It first makes sure everyone can do the familiar job. Only then

does it vary what the later handoff contains and ask how people respond when the task changes in controlled ways. A first study should preregister the familiar-competence phase, handoff contents, structural-probe definitions, calibration trials, primary outcomes, analysis family, and reduction rules. The crucial causal requirement is that routine competence be established before the handoff manipulation, so later differences cannot be blamed on unequal post-treatment practice, feedback, or extra chances to reconstruct what was missing.

7.1 First make sure everyone can do the familiar job

First, everyone has to be able to do the familiar job before anyone receives a different handoff. All participants therefore receive the same rule-only instruction and the same fixed practice dose under the original regime. Routine competence is assessed against a preregistered criterion before assignment; people who do not meet it are classified as ineligible before treatment and reported separately. Eligible participants are then randomized to one of three fixed-dose handoffs: the rule alone; the rule plus applicability or boundary information; or the rule plus applicability information and a causal or explanatory rationale. Exposure duration, presentation format, and rehearsal opportunity should be matched as closely as possible. No performance-contingent training, corrective feedback, or training-to-criterion occurs after randomization and before the critical battery. All randomized participants remain in the primary analysis regardless of any later manipulation check, and those checks should preferably occur after the critical battery so they do not become extra instruction.

7.2 Then change the world in three different ways

Then the situation changes in three different ways. Can the person avoid abandoning a good rule just because something superficial changed? Can the person notice that the old rule was never meant for this new case and withhold it without needing an unstated fallback? Can the

person remain inside the rule's domain yet re-derive an action when a parameter in the functional relation supporting the baseline rule changes? Participants encounter multiple held-out instances of those three structural probes, plus separate evidence-strength control trials. Probe membership is assigned from the task generator before outcomes are inspected. Items must include cases that warrant preservation, restriction, and modification; otherwise the battery would simply reward skepticism or rigidity. Order should be randomized or counterbalanced, and multiple surface versions per probe should keep item-specific difficulty from becoming the construct.

Confirmatory items cannot repeat examples, triggers, exceptions, critical parameter values, parameter-action pairs, or wording supplied in the handoff. A preregistered simplification test should compare the three-relation representation with a smaller preserve-versus-revise score, and the structure should contract if the simpler representation is equally informative. During the preregistered confirmatory critical battery, no correctness feedback, instructional hints, performance-contingent teaching, or other answer-revealing information is provided between trials. No trial-by-trial feedback teaches a mapping, boundary, relation, or correct action. Any later learning or feedback occurs only after confirmatory outcomes are complete or as a separately analyzed later phase.

7.2.1 Worked microdomain: Aster filtration control

A fictional task is useful here because every fact that matters can be made explicit instead of relying on a participant's outside expertise. Suppose participants operate fictional Aster filtration units. During the common baseline they learn one simple rule: for an in-domain unit, choose cartridge X. The task generator defines the domain as an aqueous feed at or below 8 L/min. The fictional rationale defines why X is adequate under the baseline configuration. Each unit has a Q-load q , and a task-defined coupling coefficient k determines capture demand D by

the relation $D = q \times k$. The available cartridges have arbitrary labels and fixed fictional capacities: $X = 8$, $Y = 18$, and $Z = 12$ capacity units. The operating principle is to choose the lowest-capacity cartridge whose capacity is at least D . Under the baseline configuration $q = 5$ and $k = 1$, so $D = 5$ and X is adequate. The rule-only handoff repeats only the action. The boundary handoff also states when the rule applies. The rationale handoff additionally states the general $D = q \times k$ relation, the three cartridge capacities, and the lowest-adequate-capacity principle. It states the baseline $k = 1$ but does not teach the critical held-out coefficient values or any corresponding cartridge choice. Casing color, serial prefix, display orientation, and plant label are declared superficial features and never enter the applicability predicate or the functional relation. All quantities and relationships in Aster are fictional task-generating facts, not claims about real filtration chemistry.

The experiment also asks a different question: can a person ignore weak reasons to abandon a good rule without becoming so rigid that strong evidence cannot move them? This is a calibration control, not a fourth handoff probe. To keep it from becoming a numeracy test, every participant learns the same evidence-state decision rule during the common pre-randomization phase: a validated evidence-state value below 0.90 means preserve the current rule while flagging uncertainty; a value at or above 0.90 means provisionally suspend or revise it. Critical control trials present the validated value directly, for example 0.20 versus 0.95; participants do not have to derive it from other quantities. Because every handoff group receives the same mapping, the design predicts no handoff-specific advantage here. Successful calibration weighs against the simplest always-preserve or always-revise accounts. Failed calibration does not uniquely diagnose rigidity, revision-proneness, numeracy, comprehension, attention, or any other latent cause.

Table 2

Held-out Aster structural probes and calibration example

Probe or control	What the participant sees	What stays fixed	Task-defined response
Surface / irrelevant change	A green-cased unit with a rotated display and new plant label remains in-domain at 6 L/min, with $q = 5$ and $k = 1$.	Applicability predicate, q , k , cartridge capacities, and the $D = q \times k$ relation all remain unchanged.	Preserve cartridge X.
Boundary change	A held-out unit has the same fictional Q-load $q = 5$ and baseline $k = 1$, but flow increases to 10 L/min.	The fictional $D = q \times k$ relation and cartridge capacities are unchanged; only the predeclared applicability predicate becomes false.	Classify the case as outside the declared domain and withhold cartridge X.

Probe or control	What the participant sees	What stays fixed	Task-defined response
Supporting- relation parameter change	An in-domain unit remains aqueous at 6 L/min with $q = 5$. A verified fictional upstream change alters the coupling coefficient from the baseline $k = 1$ to the held-out value $k = 2$. The prompt reports q and k but gives no cartridge recommendation or replacement action.	The applicability predicate, q , cartridge capacities, and the general demand principle remain fixed. What changes is the coefficient in the functional relation: the mapping from q to D changes from $D = 1q$ to $D = 2q$.	Use the rationale to compute $D = 10$ and select cartridge Z (capacity 12), the lowest- capacity cartridge that is adequate. The $k = 2 \rightarrow Z$ pair was not taught in the handoff.
Calibration: evidence strength	The unit remains an in-domain Q case with no announced structural intervention. A validated evidence-state summary is presented directly as 0.20 in one trial and 0.95 in another; the preregistered revision threshold is 0.90.	Domain variables and the supporting relation remain fixed; only the strength of the directly presented evidence state changes.	Preserve the rule while flagging uncertainty at 0.20; suspend or revise provisionally at 0.95. This calibrates response to evidence strength and does not identify a unique latent cause of failure.

Nothing about the Aster example requires a special statistical model. Before seeing any outcome, a conventional content-demand model can represent which handoff supplied the action rule, the applicability boundary, and the fictional functional rationale, as well as which relation each held-out item changes. It can include the relevant handoff-by-probe interactions, participant and item heterogeneity, surface distance, and difficulty. A diagnostic classification model could

also encode preregistered item-to-attribute requirements and multivariate response patterns when that interpretation is justified. The third probe does not escape conventional transfer merely because it is generative: a rationale advantage remains compatible with learning and applying a richer functional model. The scientific repair is narrower. The critical item is no longer a lookup of an explicitly taught condition-action pair; it requires applying a general relation to a held-out coefficient and deriving an action that was not paired with that coefficient during the handoff. The experimental design earns value only if the controlled changes reveal stable, interpretable distinctions that a simpler transfer score or probe factorization would lose.

7.3 What counts as a useful pattern

What matters is not simply how many items a participant gets right. The primary outcome is the pattern of task-defined responses across the three kinds of change: keep the rule when the relevant structure stayed intact, restrict it when the case crossed its boundary, and re-derive an action when a held-out parameter changes the functional relation supporting the baseline rule. Evidence-strength trials separately ask whether the person can preserve under weak evidence and become provisionally revisable under strong evidence. These are scoring categories, not an exhaustive psychological taxonomy. Any multivariate profile model is an implementation choice because profile-based diagnostic measurement is already established in psychometrics. Secondary measures can include how quickly the person notices a change, confidence, identification of what changed, quality of a revised rule, and transfer of the revision to another held-out case. Explanations and manipulation checks can tell us something about represented structure, but they do not prove the causal route that produced the behavior.

7.4 What the experiment has to beat

The experiment only teaches us something if it beats simpler explanations. A generic richness account predicts that giving people more information should help broadly. A conventional content-demand account already predicts selective advantages when the information in the handoff matches what a later case requires, and it may include preregistered interactions plus participant and item heterogeneity. Established diagnostic or multivariate models can analyze the same response pattern. The primary analysis must therefore be specified before the data and ask whether the three structural probes produce stable held-out distinctions that a unitary transfer score or simpler preserve-versus-revise representation would collapse. The item demands must come from the task generator, not from whichever group happens to perform differently.

The planned handoff contrasts still matter, but as tests of what information is useful under which demand, not as novelty claims. Boundary information should help when the person has to notice that the rule no longer applies. Rationale may help when the case remains in-domain but a held-out coefficient changes the functional relation that determines which action is adequate. In Aster, only the rationale handoff supplies the general $D = q \times k$ relation, the cartridge capacities, and the lowest-adequate-capacity principle; the critical $k = 2 \rightarrow Z$ pairing is never taught. A positive rationale effect would therefore show successful use of a transmitted functional model on a held-out parameter, not a newly discovered learning mechanism. Conventional content-demand transfer remains an allowed explanation. Evidence strength remains separate: everyone gets the same evidence-state mapping, so no handoff-content interaction is expected. Good control performance weighs against the simplest indiscriminate preserve/revise strategies, but a failure does not by itself explain why the participant failed.

7.5 How the proposal can fail

The design should be allowed to fail. If richer handoffs help everywhere, the result is better explained by generic information, attention, memory, or comprehension. If a conventional content-demand model captures the selective effects, that is ordinary transfer matching rather than a new statistical architecture. If the effects disappear on held-out structural cases, the participants may simply be using direct instructions. If the rule-only group performs as well as the richer groups, the extra information was unnecessary for that task. If boundary information does not improve prospective recognition of out-of-domain cases and withholding of X, the boundary component has not earned a distinct role. If the rationale condition does not outperform boundary information on held-out functional-parameter items for which the critical coefficient-action pair was never taught, rationale has not earned the proposed generative role. If performance can be reproduced by a finite lookup over conditions taught in the handoff, the third probe has failed its identification test and should contract to ordinary conditional-policy application. If the three probe relations do not produce separable held-out consequences, or if a simple preserve-versus-revise score is equally informative, the battery should contract. Evidence strength remains a separate control unless future work independently earns a handoff-specific role. And if several internal routes remain compatible with the same behavior, the result should stay at the level of behavioral dissociation.

A useful result would therefore be modest but real. The battery would show reproducible, demand-specific differences on held-out changes; successful calibration would weigh against the simplest always-preserve or always-revise strategies; and the three-relation design would retain information that a simpler score would lose. Conventional regression, multivariate, or diagnostic-classification tools may estimate that result. Such evidence would support the usefulness of this

experimental composition for asking what survived a handoff. It would not establish a universal packet of information, a unique latent causal route, a new psychometric model, or a unique explanation for calibration failures.

8. Scope and Boundaries

The paper is deliberately not saying that the person who lived through something knows better by default. It does not say people learn best through failure, that personal experience has privileged authority, that advice should be discounted, that experts should withhold conclusions, or that every inherited rule should remain perpetually open to revision. Some environments are stable; some exploration is too costly; some safety rules are appropriately conservative; and much knowledge is rationally accepted through testimony without independent reconstruction.

It also does not say that every person who fails to revise received a bad handoff. Someone may have the relevant structure available and still fail to use it because of memory, incentives, identity, social pressure, cognitive load, motivation, or institutional authority. Handoff sufficiency concerns what was made available for appropriate use and revision, not whether that information necessarily controls behavior.

Nor does the source's intention settle what the handoff did. Someone can sincerely mean to protect, teach, or help while leaving out information that later turns out to matter. That difference may be important in applied work, but this paper does not need a theory of intention to test what information survived.

Finally, advice is not the only place this problem appears. A person may inherit a policy, a routine, a technical design, a clinical rule, or a model from someone whose original learning history is unavailable. The details differ across education, organizations, engineering, clinical training, and human-AI systems. The analogy should be carried across domains only when the

same basic structure - inherited answer, changed conditions, and a need to decide what still applies - genuinely holds.

9. What This Paper Adds

If the experiment works, it provides a controlled way to test whether different handoffs support appropriate performance when later demands require keeping, limiting, or changing an inherited answer. Such patterns may show that a handoff carried information useful for those demands and may weaken simpler response accounts, but they do not identify a retained representation, comprehension state, or latent mechanism. The contribution is behavioral measurement, not a new theory of mind.

Most of the machinery around that test is already owned elsewhere. Conditional and conceptual knowledge distinguish execution from knowing when and why; transfer-appropriate processing predicts acquisition-test match (Morris et al., 1977); preparation-for-future-learning and contrasting-case work shows that routine or standard-test performance can miss differences that appear when people later have to learn or transfer (Schwartz & Bransford, 1998; Schwartz et al., 2005, 2011; Schwartz & Martin, 2004); Cognitive Flexibility Theory already targets rigidity, oversimplification, and adaptive use under novelty in complex domains (Spiro et al., 1988, 1992, 2019); explanation research already shows selective effects of causal or conceptual information (Brockbank & Walker, 2022; Cheung et al., 2019; Walker et al., 2014; Williams & Lombrozo, 2010); and diagnostic-classification psychometrics already owns item-to-attribute mappings and multivariate response profiles (Henson et al., 2009; Madison & Bradshaw, 2018; Rupp et al., 2010; Tatsuoka, 1983; Wang et al., 2026). Halbherr and Kapur's (2020) Double Treatment design also owns the higher-level experimental principle of varying both learning treatment and

assessment treatment to expose interactions that a single common test can hide. The present paper therefore does not claim a new acquisition-by-assessment architecture.

The remaining candidate contribution is narrower: a handoff-specific assessment design that establishes routine competence before treatment, varies the informational content of a compressed post-competence handoff, tests independently generated held-out cases that require preservation, boundary restriction, or re-derivation from a transmitted functional rationale under a held-out relation parameter, keeps evidence strength as a separate control, and preregisters collapse if simpler scoring or established models are equally informative. A bounded hostile search found no single prior framework clearly containing that full combination, so its status remains OPEN, not established as novel. The repaired third probe does not create a new transfer principle: conventional content-demand and Double Treatment/PFL accounts remain allowed to explain any selective advantage. If an established architecture proves scientifically equivalent, the right conclusion is a structured handoff application or synthesis. The practical lesson remains smaller than a theory and larger than a slogan: a useful handoff does not need to reproduce another person's life; it needs to support the later judgments the recipient will actually have to make.

10. Conclusion

People advance by inheriting what other people have learned. A warning, rule, procedure, or explanation can spare someone needless rediscovery, danger, and wasted effort. But compression has a cost. The answer can arrive intact while the conditions or functional relationships that made it trustworthy do not. A person can remember the words and still be poorly prepared to tell whether a new situation is merely different-looking, outside the rule's boundary, or still in-domain but governed by a changed relation.

The experiment is a way to make that hidden loss observable, not the problem itself. By first making routine performance equal and then changing different parts of the situation, it tests whether different handoffs support the appropriate response when the person should keep the inherited answer, limit it, or derive a different action from a transmitted rationale under a genuinely held-out relation parameter. Those behavioral differences can be consistent with some handoffs carrying information useful for the declared later demands, but they do not reveal a unique internal representation or causal route. That is also why correct present behavior is not enough: the important difference may be invisible until the world stops cooperating with the old rule.

The strongest version of the idea is not that experience outweighs advice or that independent judgment is superior to inherited knowledge. The scientific challenge is to understand how we can pass on what we know without forcing the next person either to rediscover it through harm or to preserve it blindly after its conditions have changed. Accumulated knowledge and continued inquiry are not opposites. The design problem is to transmit enough of the former that the latter becomes better informed rather than prematurely closed.

Funding

This work received no external funding.

Data availability

No new data were created or analyzed in this conceptual article.

Ethics statement

Not applicable. This article reports no research involving human participants or animals.

Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used OpenAI ChatGPT to support manuscript organization, literature discovery, drafting assistance, and language refinement. After using this tool/service, the author reviewed, verified, and edited the content as needed and takes full responsibility for the content of the publication.

References

- Abraham, J., Kannampallil, T. G., & Patel, V. L. (2012). Bridging gaps in handoffs: A continuity of care based approach. *Journal of Biomedical Informatics*, 45(2), 240–254. <https://doi.org/10.1016/j.jbi.2011.10.011>
- Aliaga, L., Bavolek, R. A., Cooper, B., Mariorenzi, A., Ahn, J., Kraut, A., Duong, D., Burger, C., & Gisondi, M. A. (2024). Error management training and adaptive expertise in learning computed tomography interpretation: A randomized clinical trial. *JAMA Network Open*, 7(9), e2431600. <https://doi.org/10.1001/jamanetworkopen.2024.31600>
- Audi, R. (2013). Testimony as a social foundation of knowledge. *Philosophy and Phenomenological Research*, 87(3), 507–531. <https://doi.org/10.1111/j.1933-1592.2011.00525.x>
- Barnett, S. M., & Ceci, S. J. (2002). When and where do we apply what we learn? A taxonomy for far transfer. *Psychological Bulletin*, 128(4), 612–637. <https://doi.org/10.1037/0033-2909.128.4.612>
- Bowyer, S. A., Davies, B. L., & Rodriguez y Baena, F. (2014). Active constraints/virtual fixtures: A survey. *IEEE Transactions on Robotics*, 30(1), 138–157. <https://doi.org/10.1109/TRO.2013.2283410>

- Bratthall, L., Johansson, E., & Regnell, B. (2000). Is a design rationale vital when predicting change impact? A controlled experiment on software architecture evolution. In F. Bomarius & M. Oivo (Eds.), *Product focused software process improvement* (Lecture Notes in Computer Science, Vol. 1840, pp. 126–139). Springer. https://doi.org/10.1007/978-3-540-45051-1_14
- Brockbank, E., & Walker, C. M. (2022). Explanation impacts hypothesis generation, but not evaluation, during learning. *Cognition*, 225, 105100. <https://doi.org/10.1016/j.cognition.2022.105100>
- Chase, C. C., Malkiewich, L. J., & Kumar, A. S. (2019). Learning to notice science concepts in engineering activities and transfer situations. *Science Education*, 103(2), 440–471. <https://doi.org/10.1002/sce.21496>
- Cheung, J. J. H., Kulasegaram, K. M., Woods, N. N., & Brydges, R. (2019). Why content and cognition matter: Integrating conceptual knowledge to support simulation-based procedural skills transfer. *Journal of General Internal Medicine*, 34(6), 969–977. <https://doi.org/10.1007/s11606-019-04959-y>
- Chinn, C. A., & Brewer, W. F. (1998). An empirical test of a taxonomy of responses to anomalous data in science. *Journal of Research in Science Teaching*, 35(6), 623–654. [https://doi.org/10.1002/\(SICI\)1098-2736\(199808\)35:6<623::AID-TEA3>3.0.CO;2-O](https://doi.org/10.1002/(SICI)1098-2736(199808)35:6<623::AID-TEA3>3.0.CO;2-O)
- Cohen, M. H., & Belta, C. (2023). Safe exploration in model-based reinforcement learning using control barrier functions. *Automatica*, 147, 110684. <https://doi.org/10.1016/j.automatica.2022.110684>
- Dixon, C., Oxley, E., Gellert, A. S., & Nash, H. (2023). Dynamic assessment as a predictor of reading development: A systematic review. *Reading and Writing*, 36, 673–698. <https://doi.org/10.1007/s11145-022-10312-3>
- Eiriksdottir, E., & Catrambone, R. (2011). Procedural instructions, principles, and examples: How to structure instructions for procedural tasks to enhance performance, learning, and transfer. *Human Factors*, 53(6), 749–770. <https://doi.org/10.1177/0018720811419154>
- Engel, C., Holzhausen, T., & Mischkowski, D. (2026). Trapped in the past? Routinized rule compliance and the speed of adaptation after rule change. *Judgment and Decision Making*, 21, e25. <https://doi.org/10.1017/jdm.2026.10050>
- Feldman, M. S., & Pentland, B. T. (2003). Reconceptualizing organizational routines as a source of flexibility and change. *Administrative Science Quarterly*, 48(1), 94–118. <https://doi.org/10.2307/3556620>
- Feldman, M. S., & Rafaeli, A. (2002). Organizational routines as sources of connections and understandings. *Journal of Management Studies*, 39(3), 309–331. <https://doi.org/10.1111/1467-6486.00294>
- Halbherr, T., & Kapur, M. (2020). Validation reconceived: The Double Treatment experimental design. *Frontiers in Education*, 4, 156. <https://doi.org/10.3389/feduc.2019.00156>
- Hemmerich, J. A., Van Voorhis, K., & Wiley, J. (2016). Anomalous evidence, confidence change, and theory change. *Cognitive Science*, 40(6), 1534–1560. <https://doi.org/10.1111/cogs.12289>
- Henson, R. A., Templin, J. L., & Willse, J. T. (2009). Defining a family of cognitive diagnosis models using log-linear models with latent variables. *Psychometrika*, 74(2), 191–210. <https://doi.org/10.1007/s11336-008-9089-5>
- Holyoak, K. J., & Koh, K. (1987). Surface and structural similarity in analogical transfer. *Memory & Cognition*, 15(4), 332–340. <https://doi.org/10.3758/BF03197035>
- Koedinger, K. R., & Aleven, V. (2007). Exploring the assistance dilemma in experiments with cognitive tutors. *Educational Psychology Review*, 19(3), 239–264. <https://doi.org/10.1007/s10648-007-9049-0>

- Kua, J., Lim, W.-S., Teo, W., & Edwards, R. A. (2021). A scoping review of adaptive expertise in education. *Medical Teacher*, 43(3), 347–355. <https://doi.org/10.1080/0142159X.2020.1851020>
- Loibl, K., Tillema, M., Rummel, N., & van Gog, T. (2020). The effect of contrasting cases during problem solving prior to and after instruction. *Instructional Science*, 48(2), 115–136. <https://doi.org/10.1007/s11251-020-09504-7>
- Madison, M. J., & Bradshaw, L. P. (2018). Evaluating intervention effects in a diagnostic classification model framework. *Journal of Educational Measurement*, 55(1), 32–51. <https://doi.org/10.1111/jedm.12162>
- Malfatti, F. I. (2020). Can testimony transmit understanding? *Theoria*, 86(1), 54–72. <https://doi.org/10.1111/theo.12220>
- Marchal-Crespo, L., & Reinkensmeyer, D. J. (2009). Review of control strategies for robotic movement training after neurologic injury. *Journal of NeuroEngineering and Rehabilitation*, 6, 20. <https://doi.org/10.1186/1743-0003-6-20>
- Morris, C. D., Bransford, J. D., & Franks, J. J. (1977). Levels of processing versus transfer appropriate processing. *Journal of Verbal Learning and Verbal Behavior*, 16(5), 519–533. [https://doi.org/10.1016/S0022-5371\(77\)80016-9](https://doi.org/10.1016/S0022-5371(77)80016-9)
- Paris, S. G., Lipson, M. Y., & Wixson, K. K. (1983). Becoming a strategic reader. *Contemporary Educational Psychology*, 8(3), 293–316. [https://doi.org/10.1016/0361-476X\(83\)90018-8](https://doi.org/10.1016/0361-476X(83)90018-8)
- Perry, M. (1991). Learning and transfer: Instructional conditions and conceptual change. *Cognitive Development*, 6(4), 449–468. [https://doi.org/10.1016/0885-2014\(91\)90049-J](https://doi.org/10.1016/0885-2014(91)90049-J)
- Rittle-Johnson, B., & Alibali, M. W. (1999). Conceptual and procedural knowledge of mathematics: Does one lead to the other? *Journal of Educational Psychology*, 91(1), 175–189. <https://doi.org/10.1037/0022-0663.91.1.175>
- Rupp, A. A., Templin, J., & Henson, R. A. (2010). *Diagnostic measurement: Theory, methods, and applications*. Guilford Press.
- Schwartz, D. L., & Bransford, J. D. (1998). A time for telling. *Cognition and Instruction*, 16(4), 475–522. https://doi.org/10.1207/s1532690xc1604_4
- Schwartz, D. L., Bransford, J. D., & Sears, D. (2005). Efficiency and innovation in transfer. In J. P. Mestre (Ed.), *Transfer of learning from a modern multidisciplinary perspective* (pp. 1–51). Information Age Publishing.
- Schwartz, D. L., Chase, C. C., Oppezzo, M. A., & Chin, D. B. (2011). Practicing versus inventing with contrasting cases: The effects of telling first on learning and transfer. *Journal of Educational Psychology*, 103(4), 759–775. <https://doi.org/10.1037/a0025140>
- Schwartz, D. L., & Martin, T. (2004). Inventing to prepare for future learning: The hidden efficiency of encouraging original student production in statistics instruction. *Cognition and Instruction*, 22(2), 129–184. https://doi.org/10.1207/s1532690xc12202_1
- Spiro, R. J., Coulson, R. L., Feltovich, P. J., & Anderson, D. K. (1988). *Cognitive flexibility theory: Advanced knowledge acquisition in ill-structured domains* (Technical Report No. 441). Center for the Study of Reading, University of Illinois at Urbana-Champaign. ERIC ED302821.
- Spiro, R. J., Feltovich, P. J., Coulson, R. L., Jacobson, M. J., Durgunoglu, A., Ravlin, S., & Jehng, J.-C. (1992). *Knowledge acquisition for application: Cognitive flexibility and transfer of training in ill-structured domains*. U.S. Army Research Institute for the Behavioral and Social Sciences.
- Spiro, R. J., Feltovich, P. J., Gaunt, A., & Ward, P. (2019). Cognitive flexibility theory and the accelerated development of adaptive readiness and adaptive response to novelty. In P. Ward, J. M. Schraagen, J. Gore, & E. M. Roth (Eds.), *The Oxford handbook of expertise* (pp. 951–976). Oxford University Press. <https://doi.org/10.1093/oxfordhb/9780198795872.013.41>

- Tatsuoka, K. K. (1983). Rule space: An approach for dealing with misconceptions based on item response theory. *Journal of Educational Measurement*, 20(4), 345–354. <https://doi.org/10.1111/j.1745-3984.1983.tb00212.x>
- Tullis, J. G., Goldstone, R. L., & Hanson, A. J. (2015). Scheduling scaffolding: The extent and arrangement of assistance during training impacts test performance. *Journal of Motor Behavior*, 47(5), 442–452. <https://doi.org/10.1080/00222895.2015.1008686>
- Türüç, A. (2026). Knowledge continuity at transition boundaries in digitally supported continuous improvement: A cross-case framework and the Transition Knowledge Handover Canvas. *Knowledge and Process Management*, 1–15. <https://doi.org/10.1002/kpm.70089>
- van Gog, T., Paas, F., & van Merriënboer, J. J. G. (2006). Effects of process-oriented worked examples on troubleshooting transfer performance. *Learning and Instruction*, 16(2), 154–164. <https://doi.org/10.1016/j.learninstruc.2006.02.003>
- van Gog, T., Paas, F., & van Merriënboer, J. J. G. (2008). Effects of studying sequences of process-oriented and product-oriented worked examples on troubleshooting transfer efficiency. *Learning and Instruction*, 18(3), 211–222. <https://doi.org/10.1016/j.learninstruc.2007.03.003>
- VanderStoep, S. W., & Seifert, C. M. (1994). Learning 'how' versus learning 'when': Improving transfer of problem-solving principles. *Journal of the Learning Sciences*, 3(1), 93–111. https://doi.org/10.1207/s15327809jls0301_3
- Wabersich, K. P., & Zeilinger, M. N. (2021). A predictive safety filter for learning-based control of constrained nonlinear dynamical systems. *Automatica*, 129, 109597. <https://doi.org/10.1016/j.automatica.2021.109597>
- Walker, C. M., Lombrozo, T., Legare, C. H., & Gopnik, A. (2014). Explaining prompts children to privilege inductively rich properties. *Cognition*, 133(2), 343–357. <https://doi.org/10.1016/j.cognition.2014.07.008>
- Wang, C., Quan, Y., & Arthur, D. (2026). Review of cognitive diagnostic models (CDMs): Recent methodological advancements for addressing practical challenges. *British Journal of Mathematical and Statistical Psychology*. Advance online publication. <https://doi.org/10.1111/bmsp.70066>
- Williams, J. J., & Lombrozo, T. (2010). The role of explanation in discovery and generalization: Evidence from category learning. *Cognitive Science*, 34(5), 776–806. <https://doi.org/10.1111/j.1551-6709.2010.01113.x>
- Zhang, T., Gou, J., Liu, G., Li, X., & Wei, C. (2026). Structured review practices in projects: A systems perspective on experiential knowledge transfer—A systematic review. *Systems*, 14(9), 1176. <https://doi.org/10.3390/systems14091176>
- Zwirner, E., & Thornton, A. (2015). Cognitive requirements of cumulative culture: Teaching is useful but not essential. *Scientific Reports*, 5, 16781. <https://doi.org/10.1038/srep16781>